

ORIGINAL ARTICLE

Changes in species, functional, and phylogenetic diversity of Aculeata communities along elevational gradients in the Kyushu Central Mountains of the Japanese archipelago

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Funding information

Kakenhi, Grant/Award Numbers: JP17H01912, JP19K06126; the Sasakawa Scientific Research Grant, Grant/Award Number: 2020-0524

Abstract

Climate change can degrade a community of Aculeata bees and wasps with important ecosystem roles, such as pollination and predation. In this study, we evaluated patterns in the species, functional, and phylogenetic diversity (PD) of Aculeata communities along elevational gradients in Miyazaki Prefecture, south of the Japanese archipelago. We used yellow pan traps to collect Aculeata at 200–1600 m above sea level for three seasons. We tested five functional traits (body size, trophic level, feeding guild, flight duration, and elevational range) and horizontal distribution of the species. The species diversity (SD) increased with increasing elevation. Body size, trophic level, and distribution diversity increased with increasing elevation. The community-weighted mean trophic level and flight duration decreased with increasing elevation. PD increased with increasing elevation. The number of northern species increased with increasing elevation. Trophic structure and high seasonal segregation by shortening flight duration should promote high SD at high elevation. The results of PD suggest that geographical barriers preventing the southern species migration decreased SD at low elevations. Community degradation in the lowlands can occur with geographical barriers that limit north–south migration. We show that evaluations of functional traits and phylogeny can reveal the mechanisms underlying community assembly at our sites, where both environmental and geographical conditions affect community assembly.

KEYWORDS

biogeography, climate change, community ecology, environmental filter, Hymenoptera

1 | INTRODUCTION

Aculeata bees and wasps consist of a particularly large number of species and include various feeding guilds, such as pollinators, predators, and parasitoids. They, therefore, have important ecosystem functions such as pollination, and food web control (Losey & Vaughan, 2006; Tylianakis et al., 2007). A high correlation between Aculeata species

diversity (SD) and whole arthropod SD (Duelli & Obrist, 2003) suggests that it is a useful ecological indicator (Gerlach et al., 2013). Elucidating the mechanism of community assembly in Aculeata could inform the biodiversity of various organisms.

Climate change can cause degradation of Aculeata SD and ecosystem functions. Therefore, it is necessary to understand the impact of climate change on Aculeata

community assembly. Because climate change affects the elevational range shift of species distribution, elevational gradients have been used as a model to study the effect of climate change on biological communities (Beck et al., 2010; Chen et al., 2011; Hodkinson, 2005). Because of increasing environmental stresses with increasing elevation, SD often decreases at higher altitudes (Hoiss et al., 2012; Perillo et al., 2017; Rahbek, 2005). However, the SD pattern along an elevational gradient shows different results depending on the taxon, including multiple guilds (Fontana et al., 2020). The relationship between the Aculeata community, which includes multiple guilds and elevations, is not well understood.

The factors that determine the SD in the community are niche processes and geographical dispersal restrictions (Vellend et al., 2011). Niche processes are strongly associated with the structure of functional traits (McGill et al., 2006; Webb et al., 2002), and biogeographical effects are strongly associated with phylogenetic signals. Therefore, community assembly is understood from multiple approaches to be the integration of functional traits, phylogeny, and species pool (Webb et al., 2002).

Environment selects functional traits through a niche process (Webb et al., 2002). Trait diversity and community-weighted means (CWM) are used to assess the impact of environmental constraints on community assembly (Ricotta & Moretti, 2011). Because elevational gradients create environmental stress, elevational changes are expected to affect a wide variety of functional traits (Hodkinson, 2005). Therefore, it is expected that functional traits change to stress-tolerant traits, and SD will decrease at higher elevations. For example, body size increases (Hoiss et al., 2012), and the number of generations decreases (Randall et al., 1981) with increasing elevation.

Phylogeny reflects the history of the biogeographical conditions (Tojo et al., 2017; Webb et al., 2002). Biogeographical processes that affect migration and dispersion of organisms are also important in understanding community assembly (Vellend et al., 2011; Whittaker et al., 2008). Even at the same SD, a higher phylogenetic diversity (PD) reflects more distantly related individuals within a community (Srivastava et al., 2012), which also reflects a wide range of feeding guilds. Therefore, when SD decreases due to competitive exclusion, PD does not change, because limiting similarity among relative species using similar resources may not affect the whole distant relatedness of phylogeny. Regional extinction due to geographical barriers would have occurred regardless of functional traits or feeding guilds. Geographical barriers often prevent the migration of northern species to higher altitudes and those of southern species to lower altitudes. For example, during the glacial-interglacial

cycle, warm-adapted lowland species were lost because of geographical barriers (Burwell & Nakamura, 2016).

In our study, we analyzed Aculeata bees and wasps (Hymenoptera, including Chrysidoidea, Vespoidea, and Apoidea). Although there are a few studies on a part of Aculeata along elevational gradients (Hoiss et al., 2012; Perillo et al., 2017), there have not been many studies comprehensively investigating Aculeata, including the various feeding guilds. We also investigated a wide elevational range, including warm-temperate evergreen forests and cool-temperate deciduous forests. Studies on well-developed forests are mostly conducted in tropical regions (Perillo et al., 2017), and are rare in temperate regions.

The aim of our study is to clarify the mechanism of community assembly from the diversity pattern of Aculeata along an elevational gradient within the Kyushu Central Mountains. We predict that (1) SD changes with increasing elevation, and mechanisms altering SD can be explained by functional traits and phylogenetic composition, (2) functional traits that change with elevation indicate that niche adaptation to environmental gradients may be a driver of SD, and (3) changes in PD indicate historical evolutionary processes related to migration or geographical processes may drive SD.

2 | MATERIALS AND METHODS

2.1 | Study site

The study was carried out on old secondary or natural broad-leaved forests in the Shiiba Research Forest, Kyushu University (32°21' N, 131°05' E), the Kashiba National Forest (32°21' N, 131°12' E), and the Tano Forest Science Station, Miyazaki University (31°51' N, 131°17' E) in the Kyushu Central Mountains (Figure 1). We selected 11 sites at different elevations from 177 m above sea level (a.s.l.) to 1604 m a.s.l. in these areas (Table 1). The Tano forest (177 m a.s.l.) included evergreen broad-leaved trees such as *Castanopsis sieboldii* and *C. cuspidata*. The Kashiba forest (723, 787 m a.s.l.) included evergreen broad-leaved trees such as *Quercus acta* and *Q. salicina*. In the Shiiba forest (1068, 1095, 1187, 1209, 1249, 1410, 1483, and 1604 m a.s.l.), there are deciduous broad-leaved trees, such as *Fagus crenata*, *Q. crispula*, *Acer sieboldianum*, and *Ac. pictum*, and coniferous trees, including *Abies firma*. There is little understory vegetation (such as herbs, which are pollen resources) due to overgrazing by deer in Kashiba and Shiiba. The annual mean temperature was 17.2°C at 177 m a.s.l. in Tano from 2001 to 2017, and it was 8.5°C at 1604 m a.s.l. in Shiiba from 2016 to 2017. The mean

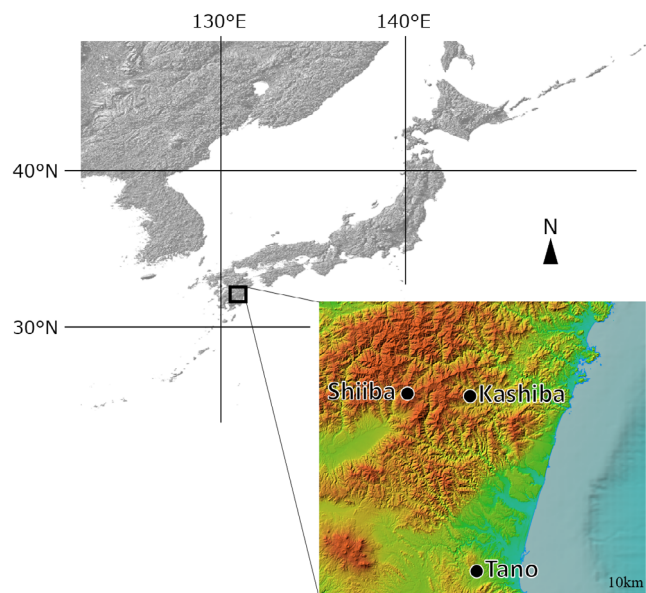


FIGURE 1 Locations of the study sites of the Shiiba Research Forest (Shiiba), the Kashiba National Forest (Kashiba), and the Tano Forest Science Station (Tano) [Color figure can be viewed at wileyonlinelibrary.com]

annual precipitation was 2950 mm in Tano from 2001 to 2017, and 3200 mm at 600 m a.s.l. in Shiiba from 2010 to 2017.

The Japanese mainland was previously connected to the Eurasian continent by a land bridge established during the glacial period (Tada, 1994). Between Kyushu, the Tsushima Islands, and the Korean Peninsula, there was a land bridge, and organisms immigrated over it (Tojo et al., 2017). In contrast, there is a sea to the south of Kyushu, of which the Tokara Strait is the biogeographical boundary that separates the Palearctic and Oriental regions (Tojo et al., 2017). This barrier restricts the migration of the warmer-adapted southern species.

2.2 | Data collection

Yellow pan traps were used for the sample collection (Figure 2). Pan traps have low human bias and are the most efficient and cost-effective methods for large-scale and long-term monitoring schemes with surveyors of different experience levels for all insect groups (Westphal et al., 2008). One limitation is that color preference differs depending on the species for all insect groups (Leong & Thorp, 1999), so that species that favor other colors may be under-sampled and their numbers underestimated. A yellow plastic bowl with a diameter of 15 cm was used. Each bowl was filled with approximately 150 ml of water with a few drops of dishwashing detergent as a surfactant.

We set up five plots under different tree canopies at each elevational site. Each plot included four yellow pans, with 20 yellow pans per elevational site. The yellow pans were set for 48 h without rain before being collected. Because the species varied depending on the season, sampling was conducted three times in different seasons: 31 March–3 June (spring), 23–26 July (summer), and 17–20 September (autumn) in 2018.

We sorted and identified Aculeata bees and wasps from the collected samples at the species level. Because our methodology was inappropriate for sampling Formicidae, they were excluded. Bee species identification followed Tadauchi and Murao (2014), and the identification of other species was mainly done by referring to Terayama and Suda (2016). The family groups followed Aguiar et al. (2013). Voucher specimens were deposited in the Entomological Laboratory, Kyusyu University, Fukuoka, Japan (ELKU collection).

2.3 | Functional traits and distribution index

We selected five functional traits that were expected to respond to elevational changes (body size, trophic level, feeding guild, flight duration, and elevational range) (Table S1). Body size was taken from the literature data of Tadauchi and Murao (2014) (for bees) and Terayama and Suda (2016) (for wasps). Trophic levels were defined at three levels: (1) herbivores (pollinator bees), (2) primary predators (predators and parasitoids of herbivores), and (3) secondary predators (predators and parasitoids of predators). Feeding guild was divided into four types that were classified according to the mechanism that adults use to feed their larvae: pollination, hunting, brood parasitism, and parasitoidism. The flight duration was calculated as the number of sampling times that the species was collected. The elevational range was calculated as the difference between the highest and lowest elevations at which the species was collected.

We also considered species distribution bias in the study. Since southern species are warm-adapted species, they are expected to be abundant in the lowlands. Since northern species are cool-adapted species, they are expected to be abundant in the highlands. We expected SD to increase when the elevational distributions of the southern and northern species overlapped. To analyze species distribution bias, we defined the distribution index of the species. The Japanese archipelago is divided into seven areas and numbered in order from the north: (1) North of Hokkaido, (2) Hokkaido, (3) East Honshu,

TABLE 1 Site description including elevation, longitude, latitude, mean annual temperature (MAT), dominant tree species, field sites, and sampling date

Elevation (m)	Longitude	Latitude	MAT (°C)	Dominant tree species	Field sites	Sampling date (all in 2018)
177	131.29803	31.86442	16.703	<i>Castanopsis sieboldii</i> , <i>C. cuspidata</i> , <i>Distylium racemosum</i> , <i>Machilus thunbergii</i> , <i>Quercus salicina</i>	Tano Forest Science Station	1–3 Jun., 24–26 Jul., 18–20 Sep.
723	131.21057	32.35313	13.427	<i>Q. acuta</i> , <i>Q. salicina</i> , <i>D. racemosum</i> , <i>Litsea coreana</i> , <i>Abies firma</i>	Kashiba National Forest	1–3 Jun., 24–26 Jul., 18–20 Sep.
787	131.20836	32.35509	13.043	<i>Q. acuta</i> , <i>Q. salicina</i> , <i>D. racemosum</i> , <i>L. coreana</i> , <i>Ab. firma</i>	Kashiba National Forest	1–3 Jun., 24–26 Jul., 18–20 Sep.
1068	131.2014	32.3503	11.357	<i>Q. acuta</i> , <i>Q. salicina</i> , <i>Acer sieboldianum</i> , <i>Ac. pictum</i>	Shiiba Research Forest	1–3 Jun., 24–26 Jul., 18–20 Sep.
1095	131.08922	32.36014	11.195	<i>Ab. firma</i> , <i>Q. crispula</i> , <i>Betula grossa</i>	Shiiba Research Forest	31 May–2 Jun., 23–25 Jul., 17–19 Sep.
1187	131.08323	32.36275	10.643	<i>Ab. firma</i> , <i>Ac. sieboldianum</i> , <i>Ac. pictum</i>	Shiiba Research Forest	31 May–2 Jun., 23–25 Jul., 17–19 Sep.
1209	131.08737	32.36806	10.511	<i>Kalopanax septemlobus</i> , <i>Cornus controversa</i> , <i>Ac. sieboldianum</i> , <i>Ac. pictum</i> , <i>Carpinus japonica</i> , <i>Ulmus laciniata</i> , <i>Pterocarya rhoifolia</i>	Shiiba Research Forest	31 May–2 Jun., 23–25 Jul., 17–19 Sep.
1249	131.07929	32.36304	10.271	<i>Fagus crenata</i> , <i>P. rhoifolia</i> , <i>Ac. sieboldianum</i>	Shiiba Research Forest	31 May–2 Jun., 23–25 Jul., 17–19 Sep.
1410	131.07762	32.36696	9.305	<i>F. crenata</i> , <i>P. rhoifolia</i> , <i>Ac. sieboldianum</i> , <i>Ab. firma</i> , <i>Car. japonica</i>	Shiiba Research Forest	31 May–2 Jun., 23–25 Jul., 17–19 Sep.
1483	131.07891	32.37204	8.867	<i>Ab. firma</i> , <i>F. crenata</i> , <i>Ac. sieboldianum</i>	Shiiba Research Forest	31 May–2 Jun., 23–25 Jul., 17–19 Sep.
1604	131.07603	32.37044	8.141	<i>F. crenata</i> , <i>Ac. Sieboldianum</i> , <i>Sorbus commixta</i> , <i>Hamamelis japonica</i> , <i>Clethra barbinervis</i> , <i>Styrax japonicus</i>	Shiiba Research Forest	31 May–2 Jun., 23–25 Jul., 17–19 Sep.

Note: All sites are natural forests. MAT at each elevation was estimated from the meteorological data of the Tano Forest Science Station (177 m a.s.l.) and the Shiiba Research Forest (600 and 1604 m a.s.l.).

(4) West Honshu and Shikoku; (5) Kyushu (including Osumi Island), (6) Amami Island and Okinawa Island, (7) Yaeyama Island and the Oriental region (Table S1). The average value in the area where a certain species was distributed was used as the species distribution index. For example, the distribution index of species distributed in Hokkaido and East Honshu was 2.5. The more southward the distribution of the species, the larger the distribution index. Distribution information for species was obtained from data from Tadauchi and Murao (2014), and Terayama and Suda (2016). An index of morphological species without a distribution record was applied in region 5 (Kyushu).

2.4 | Statistical analysis

All the samples in the five plots of 20 pans, and for the three seasons from each elevational site were pooled before conducting the analysis. For each site, the Simpson's index of SD was calculated as follows:

$$SD = 1 - \sum_{i=1}^S p_i^2, \quad (1)$$

where S is the number of species in the community, and p_i is the proportion of its species in a sample. SD was calculated using the R package “vegan” (Oksanen



FIGURE 2 A yellow pan trap. Twenty yellow pan traps were placed at each sampling elevation [Color figure can be viewed at wileyonlinelibrary.com]

et al., 2018) and “FD” (Laliberte et al., 2014; Laliberte & Legendre, 2010).

The functional diversity (FD) was calculated for different traits based on the Rao index of diversity (Rao, 1982):

$$FD = \sum_{i=1}^S \sum_{j=1}^S d_{ij} p_i p_j, \quad (2)$$

where d_{ij} expresses the dissimilarity between each trait of species i and species j . The FD index equals the sum of the dissimilarity in trait space among all possible pairs of species, weighted by the product of the species' relative abundance. The parameter d_{ij} is the Gower distance between each trait of species i and species j . Multiple FD and each trait diversity were calculated using the R package “vegan” (Oksanen et al., 2018) and “FD” (Laliberte et al., 2014; Laliberte & Legendre, 2010).

PD was calculated using the classification distance, ω , in the same way as FD, as follows:

$$PD = \sum_{i=1}^S \sum_{j=1}^S \omega_{ij} p_i p_j, \quad (3)$$

where ω was calculated using the function “taxa2dist” in the vegan package in R (Oksanen et al., 2018). Although PD is preferably analyzed using genetic information, the genetic data for most of the Aculeata in this study are not available. Therefore, the classification structure (Shimatani, 2001; Warwick & Clarke, 1995) was used to represent PD (Table S1). PD was calculated using the R package “vegan” (Oksanen et al., 2018) and “FD” (Laliberte et al., 2014; Laliberte & Legendre, 2010).

The CWM was calculated for each trait (except the feeding guild) at each site as follows:

$$CWM = \bar{t}_j = \sum_{i=1}^S p_i t_{ij}, \quad (4)$$

where t_{ij} is the trait value of trait j of species i . The CWM is weighted by the relative abundance of the species with each value. With this metric, it is possible to evaluate numerical changes such as increases or decreases in trait values with elevation by calculating the CWM at each site. The CWMs were calculated using the R package “vegan” (Oksanen et al., 2018) and “FD” (Laliberte et al., 2014; Laliberte & Legendre, 2010). The relationships between elevation and community indices, such as SD, FD, PD, and CWMs, were analyzed using a linear regression model.

The two different dissimilarities in species composition were examined to determine whether the change in species composition was “nestedness” or “turnover,” according to the difference in elevation. We partitioned the Bray-Curtis index into “nestedness” (abundance gradient, according to the explanation in the software), and “turnover” (balanced variation) components, using the R package “betapart” (Baselga et al., 2018). The distance decay of similarity was tested using the mantel test calculated using the R package “vegan” (Oksanen et al., 2018), and “ecodist” (Goslee et al., 2020). All statistical analyses were performed using R version 3.5.0, for Windows (R Core Team, 2018).

3 | RESULTS

3.1 | Species diversity

In total, we collected 918 Aculeata individuals, including 298 in spring, 289 in summer, and 331 in autumn, and 107 species, including 41 Apoidea, 39 Vespoidea, and 27 Chrysidoidea (Table S2). The most abundant species was *Anoplius pacificus* (58 individuals). *Bischoffitilla ardescens*, which was collected at 177–1483 m a.s.l., had the largest elevational range. In contrast, *Auplopus kyotensis* was only collected at 177 m a.s.l. and *Anteon jurineanum*, *A. insertum*, and *Lasioglossum zunaga* were only collected at 1604 m a.s.l. Species composition was “turnover” along the elevational distance (Table S3).

SD based on Simpson's index showed a significant linear increase with increasing elevation (Figure 3; $r = 0.800$, $p = 0.002$). Species richness also increased with elevation (Figure 4; $r = 0.663$, $p = 0.015$).

3.2 | Functional traits

Multiple functional diversities increased with increasing elevation (Figure 5, $r = 0.825$, $p = 0.001$). Our results showed significant changes in some functional traits in

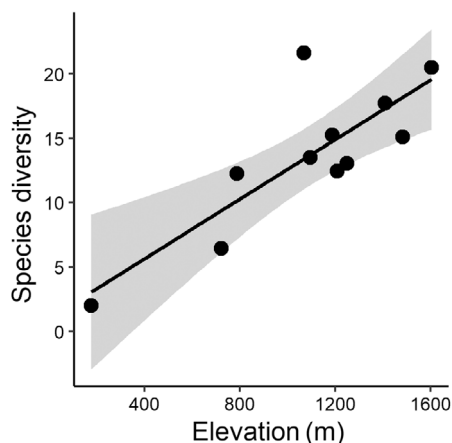


FIGURE 3 The relationship between elevation and species diversity calculated by Simpson's diversity index. The solid line and gray area indicate the result of linear regression ($r = 0.800$, $p = 0.002$) and the 95% CI, respectively

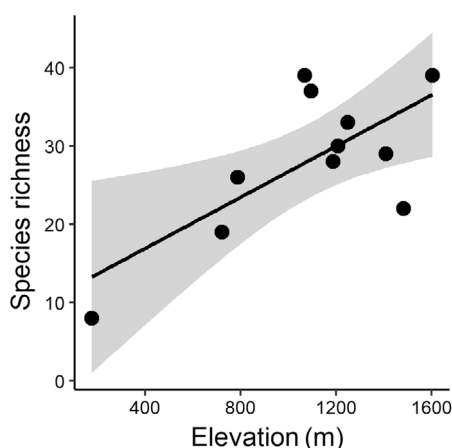


FIGURE 4 The relationship between elevation and species richness. The solid line and gray area indicate the result of linear regression ($r = 0.663$, $p = 0.015$) and the 95% CI, respectively

the Aculeata communities along an elevational gradient. The FD of trophic level and elevational range increased with increasing elevation (Table 2; $p < 0.05$). No significant changes were found in flight duration or body size diversity (Table 2; $p > 0.05$).

Trophic level CWM decreased with increasing elevation (Table 2; $p < 0.05$). The CWM of flight duration also decreased with increasing elevation (Table 2; $p < 0.05$). The relative abundance of pollinators (herbivores) increased, and the relative abundance of predators decreased with increasing elevation (Figure 6). The relationships between the CWMs of body size, elevation range, and elevation were not statistically significant (Table 2; $p > 0.05$).

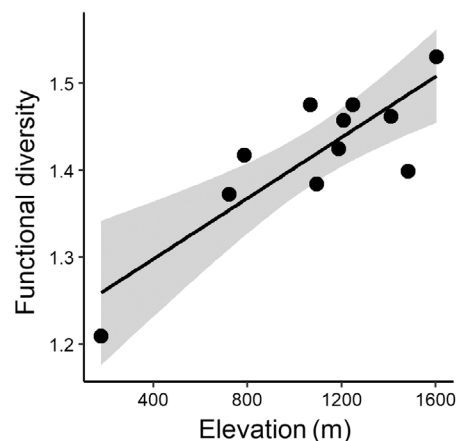


FIGURE 5 The relationship between elevation and multiple functional diversity calculated by Rao's quadratic entropy. The solid line and gray area indicate the result of linear regression ($r = 0.825$, $p = 0.001$) and the 95% CI, respectively

TABLE 2 Linear regression's correlation coefficient between elevation and community-weighted mean (CWM) or functional diversity (FD) of each functional trait

Functional traits and distribution index	CWM (r)	FD (r)
Trophic level	−0.735**	0.781**
Flight duration	−0.632*	0.0458
Mean body size	−0.0949	0.310
Altitudinal range	−0.387	0.640*
Distribution index	−0.906***	0.883***

* $p \leq 0.05$.

** $p \leq 0.01$.

*** $p \leq 0.001$.

3.3 | Phylogenetic composition

Our results showed significant changes in the phylogenetic composition of the Aculeata communities along an elevational gradient. The PD increased with increasing elevation (Figure 7; $r = 0.819$, $p = 0.001$). Distribution index diversity increased with increasing elevation (Table 2; $p < 0.001$). The CWM of the distribution index decreased with increasing elevation (Table 2; $p < 0.001$); that is, the number of northern species increased with increasing elevation.

All results of diversity and CWM for each elevation showed in Table S4.

4 | DISCUSSION

Aculeata community species and functional and phylogenetical compositions respond to elevational gradient,

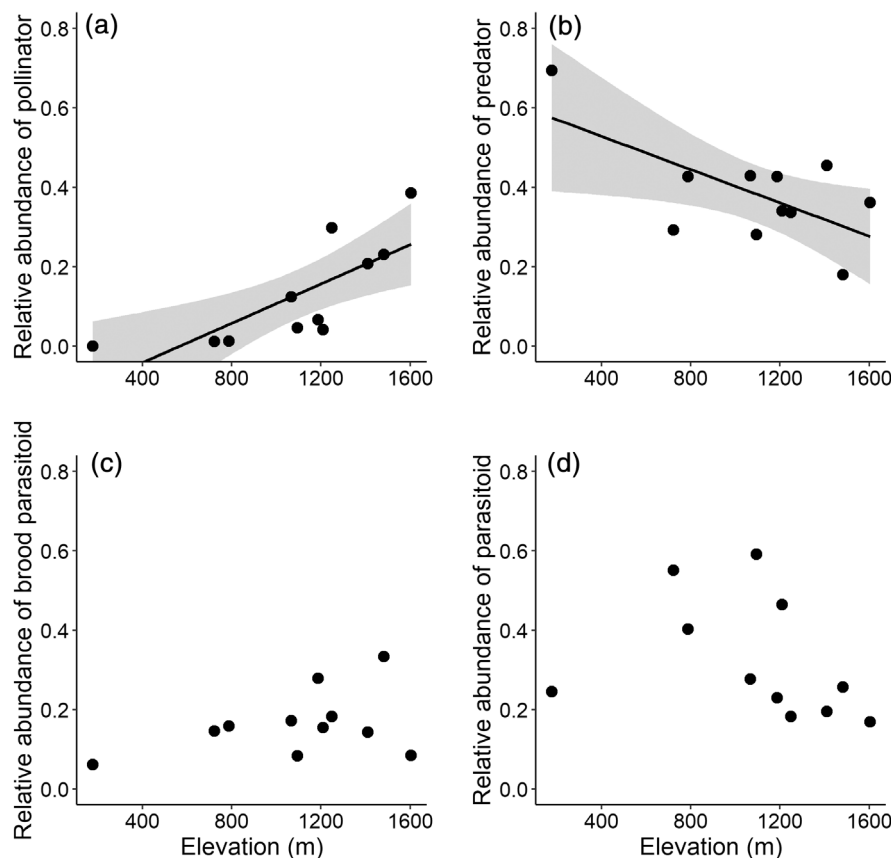


FIGURE 6 The relationship between elevation and relative abundance of each feeding guild. The solid line and gray area indicate the result of linear regression and the 95% CI, respectively. (a) Pollinator ($r = 0.721$, $p = 0.007$), (b) predator ($r = -0.592$, $p = 0.03$), (c) brood parasitoid ($p > 0.05$), and (d) parasitoid ($p > 0.05$)

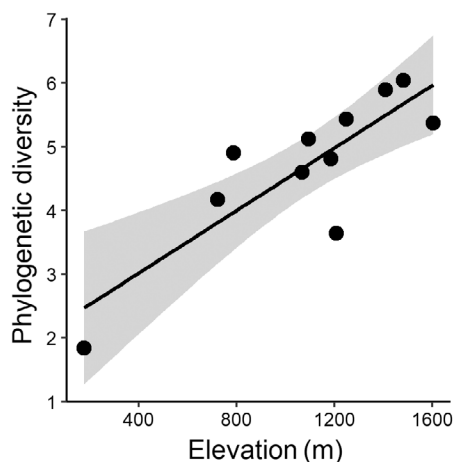


FIGURE 7 The relationship between elevation and phylogenetic diversity. The solid line and gray area indicate the result of linear regression ($r = 0.819$, $p = 0.001$) and the 95% CI, respectively

indicating that both the niche process and evolutionary processes contribute to Aculeata SD with elevation in Kyushu Central Mountain. This study showed that trophic structure and high temporal resource partitioning by shortening flight duration should result in high SD at high elevation. Flight duration time among northern species is generally shorter than among southern species,

suggesting distribution biases and biogeographic conditions may also correlate with SD at high elevations. Indeed, SD and PD decrease at low elevations, indicating that immigration restriction is reinforced by geographical barriers. This study shows that the relationship between functional traits and temporal segregation, together with phylogenetic background and biogeographical conditions, is an important contributor to the SD of the Japanese Aculeata assembly.

Increases in FD of several traits at high elevations indicate that species adopt different strategies and functional traits to coexist. Our research site on Kyushu Island has a warm, humid environment, with an annual mean temperature of 8.5°C at the highest elevation (1604 m a.s.l.). Because physical stresses on Aculeata communities at our sites are mild, SD might not decrease at high elevations as a result.

This study showed that the CWM of flight duration, in correlation with seasonal activity range or voltinism, was a more responsive functional trait to environmental change than body size or elevation range, contrary to previous reports (Hoiss et al., 2012; Rapoport, 1982). This suggests that seasonal segregation occurs at high elevations, whereas species with longer flight durations dominate at low elevations, because of the physical stress gradient. Climate change is promoting an increase in the

total number of reproductive cycles per year, with consequent population growth, and an expanding distribution whereas the population of univoltine habitat specialists is decreasing (MacGregor et al., 2019). In addition, it is known that the number of generations (comparable to flight duration) of flies is significantly reduced at high elevations in northern England (Randall et al., 1981). These suggest that climate change will promote population expansions among multivoltine species, but disfavor species with short flight durations (univoltine) that inhabit high elevations. *A. kyotensis*, a common species at the lowest elevation, and *Tiphia sternata*, *Clistoderes futabae*, and *Caliadurgus ussuriensis*, which occur at low to high elevations, appeared at all sampling times.

The CWM of trophic levels indicates that environmental stress increases with increasing elevation caused trophic levels to shrink. In addition, there were few pollinators at low elevations. Pollen resources on forest floors that are accessible to bees may be plentiful at high elevations, because forest floors are brighter in high mountain deciduous forests than in evergreen forests at low elevations. At our site, dwarf bamboo (*Sasa borealis*) once dominated, but now there is little understory vegetation due to deer overgrazing. The abundant pollinators in the highlands probably use pollen resources provided by trees that are enriched at higher elevations. Besides, synchrony between pollinators and flowers can easily be lost because bee phenology is less sensitive to climate change than flower phenology (Stemkovski et al., 2020). In contrast, generalist predators are expected to become extinct less quickly because of their wide host range (Damien & Tougeron, 2019). This suggests that warm-adapted pollinators were more likely to become extinct due to geographical barriers during the Last Glacial Maximum (LGM) than were predators on Kyushu Island.

The negative relationship between the CWM of distribution index and elevation, and the positive relationship between diversity of distribution index and elevation indicated that south-species were not excluded by cool climate at high elevation within this study area. Our results of significant changes in PD indicated that low SD is not due to competitive exclusion at low elevations. Therefore, low PD at low elevations implies that warm-adapted species were bottlenecked by geographical barriers in the past, and species have not yet been replenished. The Tokara Gap between the Kyushu and Ryukyu Islands is a significant global biogeographical boundary between the Oriental and Palearctic regions (Tojo et al., 2017). It is postulated that the southern species that were less tolerant were driven further south and became locally extinct, and they could not return, although the land bridges during the LGM contributed to

the coming and going of the northern, cold-tolerant species. As some plants (Momohara, 2008) and large mammals (Iwase et al., 2012) disappeared from Japan because of cold climates during the LGM, we speculate that this climatic change also affected Aculeata communities. This phenomenon is not unique to Japan but was also found in Australia (Burwell & Nakamura, 2016). Low ant richness at low elevations suggests that it reflects historical biotic attrition owing to a lack of lowland glacial refugia (Burwell & Nakamura, 2016). In addition, plant richness in Western Europe is low owing to geographical barriers such as the Alps (Davis, 1983). These results indicate that the lack of glacial refugia and restricted migration from low latitudes should reduce SD in the lowlands. In contrast, northern, cold-tolerant species could survive at low elevations during the LGM and could escape to high elevations as the climate became warmer. Their diversity was not reduced owing to the lack of glacial refugia and migration restrictions.

Overall, we found that the pattern of SD increased with increasing elevation in the Aculeata community in the Kyushu Central Mountains. We also showed that species coexistence imposed by seasonal segregation increased SD at high elevations, and restricting immigration of warm-adapted species by geographical barriers decreased SD at low elevations. Minor degradation of communities at low elevations can occur with biogeographical barriers that limit north–south migration. We highlight the importance of evaluating the geographical background to reveal the mechanism of Aculeata diversity.

ACKNOWLEDGMENTS

We thank the Shiiba Research Forest, Kyushu University, the Kashiba National Forest, and the Tano Forest Science Station, Miyazaki University, for permitting sampling, and the Shiiba Research Forest and the Tano Forest Science Station for weather data. We also thank Naoaki Tashiro for surveying and identifying the dominant tree species in Shiiba, and Ryuji Ichihashi, Kiyohiro Kaji, Daisuke Ogi, Keiichiro Cho, Shusuke Murata, Masaki Akesaka, and Erika Kawakami. We would like to thank Editage (www.editage.com) for English language editing. This study was partly supported by “Kakenhi” grants (Nos. JP17H01912 and JP19K06126) from the JSPS KAKENHI, and by the Sasakawa Scientific Research Grant (No. 2020-5024) from the Japan Science Society.

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How to cite this article: Uemori, K., Mita, T., & Hishi, T. (2021). Changes in species, functional, and phylogenetic diversity of Aculeata communities along elevational gradients in the Kyushu Central Mountains of the Japanese archipelago. *Ecological Research*, 36(5), 778–787. <https://doi.org/10.1111/1440-1703.12242>